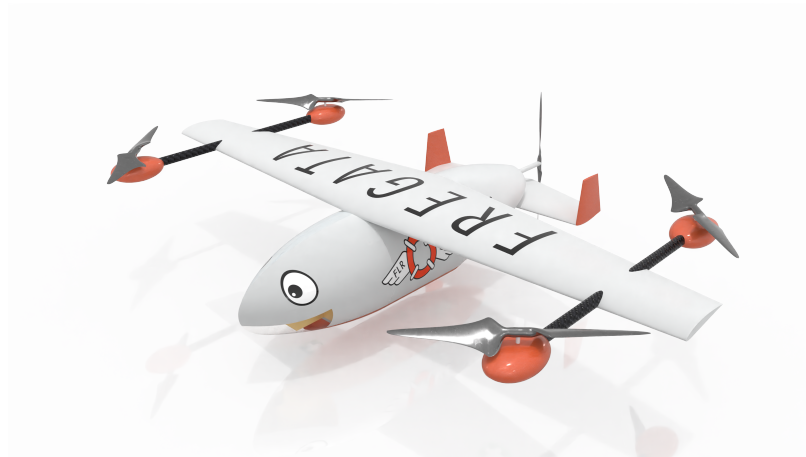




Flying Liftercraft

FLR05 Aileron, Control Derivatives and Autopilot Design



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Abstract

This report mainly focus on the fixed wing mode of the Flying Liferfts 22 UAV, which consist three parts: aileron design, control derivatives and autopilot design. In aileron design, ailerons will be preliminary sized, and validate its performance to meet mission requirement. and it is designed to extend from from 40.91% of span to 63.6%, with chord ratio 0.3 and max deflection $\pm 15^\circ$. Then the control derivatives for aileron, rudder, elevator and throttle could be calculated based on the design. Finally, lateral and longitudinal model could be set up to design autopilot, PID controller is used for roll, yaw, climb gradient and velocity control, where all the control gains are calculated at the end

Contents

List of Figures	iv
List of Tables	v
1 Introduction	1
2 Aileron Design	1
2.1 Preliminary Sizing	1
2.2 performance Estimation	1
2.3 Other Aileron Design Constrains	3
2.3.1 Aileron Reversal	3
2.3.2 Adverse Yaw	3
2.4 Final Design Decision	3
3 Control Derivatives	3
3.1 Aileron Control Derivatives	4
3.2 Elevator Control Derivatives	5
3.3 Throttle Control Derivatives	5
3.4 Rudder Control Derivatives	6
3.5 Dimensional and Final Control Derivatives	6
4 Model Set Up and Autopilot Design	7
4.1 Lateral Model	7
4.2 Lateral Control performance	7
4.3 Longitudinal Model	9
4.4 Longitudinal Control performance	9
4.5 Final Control Gains	11
5 Conclusions and Futher Work Recommendations	11

List of Figures

1	Performance requirement	1
2	Aileron Effectiveness Parameter vs Chord Ratio	2
3	Aileron Performance	3
4	Yawing moment coefficient vs lift coefficient	4
5	Lateral Model in Simulink	7
6	Comparison of Rolling Performance with and without Roll Rate Control	8
7	Aircraft Performance with Controller for a 30° Coll by Simulink	8
8	Aircraft Rolling Performance by X-Plane	8
9	Longitudinal Model	9
10	Performance of Climb Gradient at 10°	10
11	performance of Climb gradient with X-Plane	10
12	Performance of Velocity Controller from 40m/s to 58.3m/s	10
	Conventional Factors for Aerodynamics Derivatives	13

List of Tables

1	Aileron Parameters	3
2	Control Derivatives Values	6
3	Control Derivatives Values	11

1 Introduction

The following report describes the design of aileron, calculation of control derivatives and autopilot controller design of a Flying Liferaft. These are the main design points:

Preliminary sizing the aileron, with iteration of its mission requirement and constraints, find a estimated performance

Calculating aileron, rudder, elevator and throttle control derivatives, with this specific design

Build state space model based on the UAV equation of motion, find controller to ensure stable flight and efficient maneuver.

2 Aileron Design

2.1 Preliminary Sizing

Aileron is an important control surface for the aircraft roll control performance. To size the aileron, following parameters could be determined to make a start.

The aileron span is designed from $0.45m$ to $0.7m$ (from 40.91% to 63.6%) along the wing. The main constrains of inboard and outboard position in this design are as follows:

1. The stall position is at $0.45m$ along wings span, to prevent stalling at aileron and avoid loss of performance, the inboard position is constrained.
2. The pylons connecting rotors and wing is mounted inside the wing at $0.79m$, in which the outboard position should be less than that to avoid disturbance created by pylons.

The chord ratio $\frac{C_a}{C} = 0.3$ in which the aileron will connect with rear spar. In this case, the rear bar can act as an attachment point to reduce overall structure weight.

The maximum deflection angle δ_{Amax} was set to be at $\pm 20^\circ$ the beginning, which could prevent aileron stall while maximize its performance. As stated by Sadraey [1], flow separation tends to occur when aileron deflects more than about 20° to 25° . However, with further calculation and simulation $\pm 15^\circ$ was proven to be sufficient for the UAV.

In terms of the control performance needed to be achieved, the following figure from Sadraey [1] suggested that this aircraft should achieve a desired bank angle Φ_{des} of 30° within 1.3 seconds. As the performance of the UAV in cruising stage was regarded as Class I with flight phase C

(a) Time to achieve a specified bank angle change for Class I			
Level	Flight phase category		
	A	B	C
	Time to achieve a bank angle of 60° (s)	Time to achieve a bank angle of 45° (s)	Time to achieve a bank angle of 30° (s)
1	1.3	1.7	1.3
2	1.7	2.5	1.8
3	2.6	3.4	2.6

Figure 1: Performance requirement

2.2 performance Estimation

In this section, the performance of the designed aileron, specifically its response time, could be calculated to ensure the design meet the mission requirements.

At first, to determine rolling moment coefficient C_l , the aileron roll control power $C_{l_{\delta A}}$ should be calculated in advance, which could refer to the following equation based on literature [1]

$$C_{l_{\delta A}} = \frac{2C_{l_{\alpha w}}\tau C_r}{S_{ref}} \left[\frac{y^2}{2} + \frac{2}{3} \left(\frac{\lambda - 1}{b} \right) y^3 \right]_{y_i}^{y_o} \quad (1)$$

Where τ is the aileron effectiveness parameter and is obtained from Figure 2 below, y_i and y_o is the inboard and outboard position, while $C_{l_{\alpha W}}$ is the 3D wing lift curve slope in the aileron section. In this case, $C_{l_{\alpha W}}$ is assumed to be the same as wing lift curve slope, which is $4.6338rad^{-1}$. As a result, C_l could be determined using the following formula:

$$C_l = C_{l_{\delta A}} \delta_A \quad (2)$$

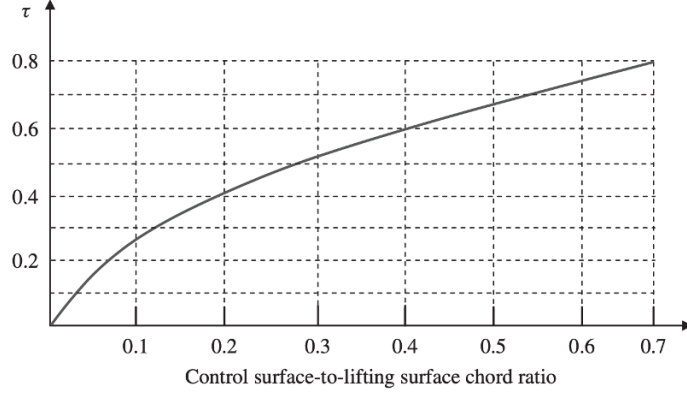


Figure 2: Aileron Effectiveness Parameter vs Chord Ratio

Then the aerodynamics rolling moment L_A can be modeled using the formula below:

$$L_A = \bar{q} S C_l b \quad (3)$$

Where $\bar{q}$ is dynamic stability.

In the next step, steady-state roll rate P_{ss} could be found in determine time required to achieve a desired bank angle. Considering that when the UAV is rolling with a constant roll rate, the rolling moment produced by aileron is the same as that of drag, the steady-state roll rate P_{ss} can be calculated as follows:

$$P_{ss} = \sqrt{\frac{2L_A}{\rho (S_{ref} + S_h + S_{vt}) \cdot C_{D_R} \cdot y_D^3}} \quad (4)$$

Where S_h and S_{vt} are area of horizontal and vertical tailplane; y_D average of three control surface distance, which is assumed to be at 40% of wingspan [2]; C_{D_R} is rolling drag coefficient usually lies between 0.7 and 1.2, which is assumed to be 1.1 in this design as four rotors for VTOL would produce larger drag than usual.

The bank angle Φ_1 when it reaches steady state roll rate can then be determined:

$$\Phi_1 = \frac{I_{xx}}{\rho (S_{ref} + S_h + S_{vt}) \cdot C_{D_R} \cdot y_D^3} \ln(P_{ss}^2) \quad (5)$$

Then the rate of roll rate $\dot{P}$ can be calculated:

$$\dot{P} = \frac{P_{ss}^2}{2\Phi_1} \quad (6)$$

Finally, the time required to achieve a turning angle could be calculated:

$$t_{req} = \sqrt{\frac{2\Phi_{des}}{\dot{P}}} \quad (7)$$

If t_{req} is less than 1.3s, it is then shows that the design of aileron would satisfy rolling control requirement. Figure 3 shows the rolling performance of the designed aileron. In this design specifically, the aircraft can achieve a 30° roll within 0.95s through calculation, which proved the design can meet the mission requirement.

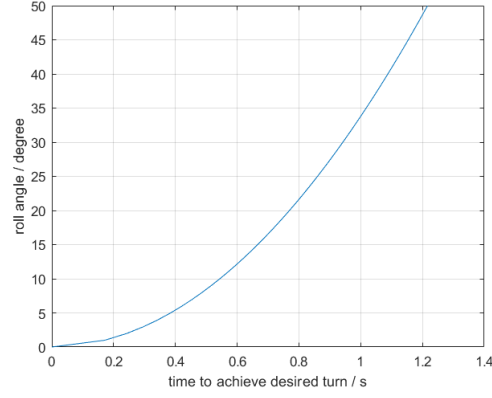


Figure 3: Aileron Performance

2.3 Other Aileron Design Constrains

2.3.1 Aileron Reversal

Due to aeroelastic effect, ailerons might produce reversed rolling moment especially in supersonic flight. However, the UAV is supposed to fly in a cruise speed at $58.3m/s$, which is relative low, so the effect of aeroelastic can be ignored in this design.

2.3.2 Adverse Yaw

Another constrain needed to be considered is adverse yaw, it would occur mostly at low speed. As the aileron deflects, the aileron which deflects downwards would produce more lift and drag, which will lead to an adverse yaw. To address this, a simultaneous rudder deflection could be applied when rolling. More details of addressing adverse yaws will be discussed in later sections.

2.4 Final Design Decision

After iteration, the final design parameters are as follows:

Table 1: Aileron Parameters	
Design Parameters	Values
Inboard Position (%)	40.9
Outboard Position (%)	63.6
Chord Ratio (%)	30
Max Deflection ($^{\circ}$)	± 15
Area (m^2)	0.0472

3 Control Derivatives

To design an autopilot for this aircraft, the performance of control surfaces and throttle should be modified into the equation of motion, and control derivatives was calculated. From lecture notes [2], the symmetric and asymmetric states are completely independent of each other. As a result, the control derivatives matrices for longitudinal and lateral control input were shown as follows:

$$B_S = \begin{bmatrix} X_{\delta_E} & X_{C_T} \\ Z_{\delta_E} & Z_{C_T} \\ M_{\delta_E} & M_{C_T} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B_A = \begin{bmatrix} Y_{\delta_A} & Y_{\delta_R} \\ L_{\delta_A} & L_{\delta_R} \\ N_{\delta_A} & N_{\delta_R} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (8)$$

The rest of the section will focus on calculating all these derivatives based on the specific design of the UAV.

3.1 Aileron Control Derivatives

The rolling moment derivatives due to aileron at subsonic speed L_a could be determined using the method described in ESDU 88013 [3], in which the formula is shown as follows:

$$L_a = -\frac{1}{2}\bar{\eta}\frac{\partial C_L}{\partial \delta}(\Phi_{ai} - \Phi_{ao}) \quad (9)$$

Where $\bar{\eta}$ is moment arm; Φ_{ai} and Φ_{ao} are inboard and outboard limit of the aileron, found using empirical data; $\frac{\partial C_L}{\partial \delta}$ is the rate of change of lift coefficient due to aileron deflection, can be calculated using equation below:

$$\frac{\partial C_L}{\partial \delta} = \left[\frac{\partial C_L}{\partial \delta} / \frac{\partial C_L}{\partial \alpha} \right]_T \left[\frac{\partial C_L}{\partial \alpha} \right]_T [1 - k_1 k_2] \quad (10)$$

Where $\frac{\partial C_L}{\partial \alpha}$ is the lift curve slope, factor k_1 and k_2 can also from empirical graphs.

With above calculation, the non-dimensional rolling moment derivative due to aileron is equal to -0.2081 rad^{-1}

The yawing moment coefficients due to aileron can also be determined using ESDU 88029 [4]. The coefficient C_n is the sum of induced drag C_{ni} and profile drag C_{np} components, where C_{np} is a constant changing with deflection of ailerons

$$C_{np} = (\eta_i - \eta_o) \frac{\bar{\eta}}{4} \frac{C_a}{C} \cos \Lambda_{\frac{1}{4}} \cos^2 \Lambda_h (D_s - D_p) \quad (11)$$

$$D_n = \left[1 - (0.05\gamma' \cos \Lambda_h)^2 \left(1 - \frac{|\xi'_n + \gamma'|}{\xi'_n + \gamma'} \right) \right] \left(\frac{\xi'_n + \gamma'}{57.3} \right)^2 \quad (12)$$

Where ξ'_s is starboard deflection, positive trailing edge down; ξ'_p is that of port, negative trailing edge up. It clear shows that when deflection angle on both sides are equal, C_{np} is equal to zero.

With no flaps, C_{ni} is a function linearly about C_L , which can be calculated as follows:

$$C_{ni} = -GC_L L_a \xi' + \frac{H}{AR} \left(\frac{1}{2} (\xi'_p - \xi'_s) \cos \Lambda_h - J_\delta \delta_t \right) L_a \xi' \quad (13)$$

Where G , H and J_δ are empirical parameters, δ_t is the twist angle of wing from root to tip. After adding two components C_{ni} and C_{np} , a figure of C_n vs C_L when both aileron deflection angles are all equal to 15° can be plotted below:

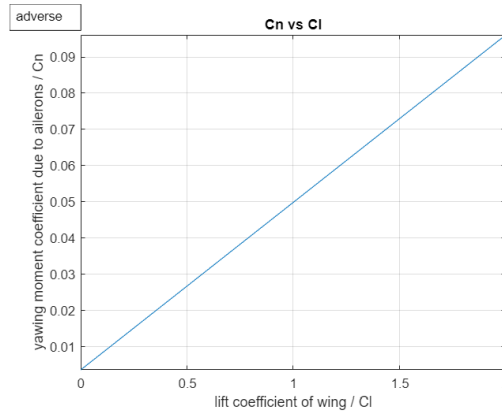


Figure 4: Yawing moment coefficient vs lift coefficient

It shows that in most of the cases when both aileron deflection angle is the same, there would be adverse yaw. With data from Aerodynamic Team, C_l at cruise is 0.3467, so the yawing moment coefficient C_n is equal to 0.02 when cruising. From Babister, A. W. [5], non-dimensional yawing moment coefficient is equal to yawing moment control derivative, which means N_a is also equal to 0.02.

By comparing the magnitude of yawing moment and rolling moment produced by aileron, the yawing effect is relatively small. There are two ways to encounter this problem.

First, a different angle of aileron deflection could be used to adjust value of C_{np} and make yawing moment to be zero at different lift coefficient. For example, when cruising and C_L is equal to 0.4, the deflection of starboard aileron should be 18° and port 20° to eliminate adverse yaw in an anticlockwise roll. However, it is different in this stage to calculate difference in deflection under each situation. As a result, the second method is applied, which is using rudder to encounter the adverse yaw. An adverse yaw controller is applied, once there is an aileron deflection to produce an instantaneous rudder deflection to encounter adverse yaw. More details of the controller would be described later in this report

Finally, the sideforce derivatives due to aileron can be defined using moment of equilibrium:

$$Y_a = N_a / \bar{\eta} \quad (14)$$

Equal to 0.0348 in cruising.

3.2 Elevator Control Derivatives

For elevator deflection derivatives, the following equation from the note [2] could be used to find Z derivatives:

$$Z_{\delta_E} = -\frac{1}{2} \rho U_e^2 S_H a_E \quad (15)$$

Where a_E is the lift curve slope of elevator, which is estimated to be the same for that of horizontal tailplane.

However, for X derivatives, at first X force could be found using the formula below:

$$X_E^a = -\frac{1}{2} \rho U_e^2 S_H C_{D_E} \quad (16)$$

Where C_{D_E} is the sum of C_{D0_E} and $C_{L_E}^2 / (\pi A R_E e_E)$.

By taking the derivatives, and given that $\frac{C_{L_E}}{\partial \delta_a}$ is equal to a_E , the X force derivatives can be calculated as follows:

$$X_{\delta_E} = -\rho U_e^2 S_H a_E / (\pi A R_E e_E) \quad (17)$$

Where Oswald efficiency is determined to be zero in this case.

Through taking moment about centre of gravity, M_{δ_E} can be found

$$M_{\delta_E} = (x_E - x_{cg}) Z_{\delta_E} - (z_E - z_{cg}) X_{\delta_E} \quad (18)$$

3.3 Throttle Control Derivatives

The throttle control derivatives X_T , Z_T and M_T could be found using the following methods:

The force in X direction could be found:

$$X = T \cos(\epsilon + \alpha) \quad (19)$$

So, the derivative with respect to thrust T can be calculated:

$$X_T = \cos(\epsilon + \alpha_e) \quad (20)$$

Where ϵ is the thrust setting angle and α_e is the angle of attack when cruising. With similar approach, Z_T is also calculated:

$$Z_T = -\sin(\epsilon + \alpha_e) \quad (21)$$

For pitching moment derivatives by throttle, it could be found by taking the moment equilibrium about center of gravity. From notes, total moment neglecting elevator and trim tab can be found:

$$M^a = T l_T + \frac{1}{2} \rho U_e^2 S_{ref} \bar{c} (C_{M0} - K_n C_l) \quad (22)$$

By taking derivatives, M_T could be found:

$$M_T = l_T = (X_H - X_T) \sin(\epsilon + \alpha_e) + (Z_H - Z_T) \cos(\epsilon + \alpha_e) \quad (23)$$

As a result, the value of non-dimensional thrust control derivatives X_T , Z_T and M_T is calculated to be 0.9997, -0.0253 and -0.0139.

3.4 Rudder Control Derivatives

The rudder control derivatives is calculated using the method stated in ESDU 87008 [6], non-dimensional sideforce derivatives Y_r can be calculated using as follows:

$$Y_r = -(Y_v)_{FR} \alpha_\delta \Delta\Phi \quad (24)$$

Where $(Y_v)_{FR}$ is the sideforce derivative due to sideslip, α_δ is control effectiveness parameter and $\Delta\Phi$ is part-span correction parameter. $(Y_v)_{FR}$ can be modified as follows:

$$(Y_v)_{FR} = -J_R J_T a_{1F} S_F / S_W \quad (25)$$

Where J_R and J_T are factors relating the geometry of tailplanes, especially J_T has been taken into account that the endplate effect would increase effective horizontal tailplane span; a_{1F} is lift curve slope, which is estimated from ESDU 70011 [7].

The parameter α_δ was calculated as follows:

$$\alpha_\delta = (\alpha_\delta)_{th} (1 - k_1 k_2) \quad (26)$$

k_1 and k_2 are correction factor from empirical data based on geometry, and $(\alpha_\delta)_{th}$ is the theoretical value effectiveness. In order to define $(\alpha_\delta)_{th}$, the equivalent aspect ratio for the vertical tailplane A_{Feq} should be found first. And A_{Feq} can also be found from empirical graph with geometry and $(Y_v)_{FR}$. Together with rudders' chord ratio C_r/C_v , $(\alpha_\delta)_{th}$ was found, so as α_δ . After that, $\Delta\Phi$ was calculated using the following formula:

$$\Delta\Phi = \Phi_2(\eta_o) - \Phi_2(\eta_i) \quad (27)$$

Where Φ is empirical correction factor based on inboard and outboard position of rudders. With all parameters calculated, non-dimensional sideforce derivatives due to rudder Y_r is equal to 0.1111.

With the sideforce calculated, the yawing and rolling moment could be calculated by taking the moment equilibrium about centre of gravity as shown below:

$$N_r = -Y_r (l_R \cos \alpha + z_R \sin \alpha) / b \quad (28)$$

$$L_r = Y_r (z_R \cos \alpha - l_R \sin \alpha) / b \quad (29)$$

Where l_R and z_R are vertical and horizontal with respect to cg, measured from aerodynamics centre of vertical tailplane.

Finally, l_R and z_R are equals to -0.0912 and 0.0169 separately.

3.5 Dimensional and Final Control Derivatives

After calculating all the control derivatives, the rudder, aileron and throttle derivatives are still non dimensional. Conversion factors should be applied to those derivatives, the factors are stated by ESDU 86021[8], and it is shown in Appendix.

Finally, all the values of all the dimensional control derivatives are shown in the table below:

	Lateral		Longitudinal
Y_{δ_A}	112.90	X_{δ_E}	-6.2674
L_{δ_A}	-675.34	Z_{δ_E}	-296.69
N_{δ_A}	64.917	M_{δ_E}	-163.06
Y_{δ_R}	20.462	X_{C_T}	1427.1
L_{δ_R}	-3.9718	Z_{C_T}	-36.117
N_{δ_R}	0.7373	M_{C_T}	-28.979

Table 2: Control Derivatives Values

4 Model Set Up and Autopilot Design

4.1 Lateral Model

In order to test the open loop response in different dynamic modes and find controller for roll and yaw, a simulation was built in Simulink, which is shown in the Figure 5 below.

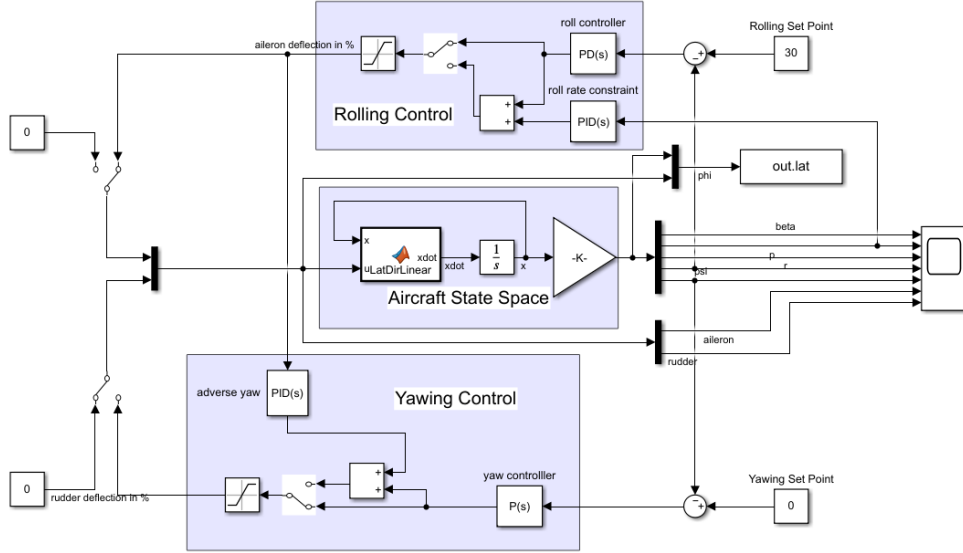


Figure 5: Lateral Model in Simulink

Where the asymmetric states was set in the middle, with integration, the states of the aircraft x can be calculated at each time step, they are sideslip velocity, roll rate, yaw rate. roll angle and yaw angle. The input u are aileron and rudder deflection as percentage, which will be converted into deflection in radians in the state space block.

4.2 Lateral Control performance

Two controller of roll and yaw are involved in lateral control. PID controllers is selected for all the controller in both lateral and longitudinal models, as it is easy to implement and easy to tune with. The rolling controller consist a roll angle controller and roll rate controller.

The roll angle controller is main control input, where the difference between rolling set point and UAV roll angle is calculated and conduct into the controller. When tuning the controller, as the rolling is slightly responsive, an small P gain is design with D gain to encounter overshoot and settling time. I term is not considered in this case. That is because I term increase overshoot given that it is the main problem with a too responsive control input, and the steady state error very small.

The roll rate P controller may also be added in rolling control to stabilise roll rate. The rolling performance of the aircraft with and without roll rate control is shown below:

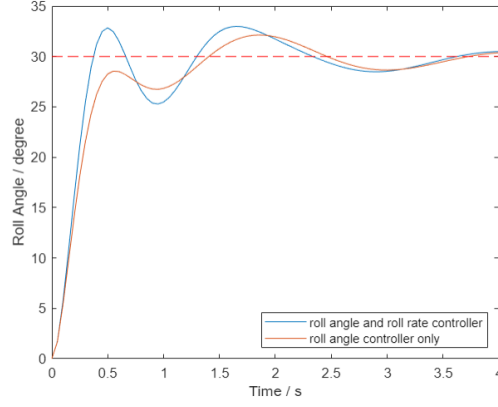


Figure 6: Comparison of Rolling Performance with and without Roll Rate Control

It can be found that the extra roll rate controller is an external P controller could slightly decrease the rise time. However, given the performance without roll rate control is already fast and stable, roll rate controller is not necessary in this design.

The roll controller also need to encounter the torque generated by back pusher. Through calculation of rolling moment balance, the aileron should have a 2° deflection when cruising.

At the end, the roll controller worked precise and fast in both Simulink and X-Plane, which could achieve a 30° roll with $0.5s$ rise time, 3° overshoot and $4s$ settling time. The Performance from Simulink and X-Plane testing is shown below:

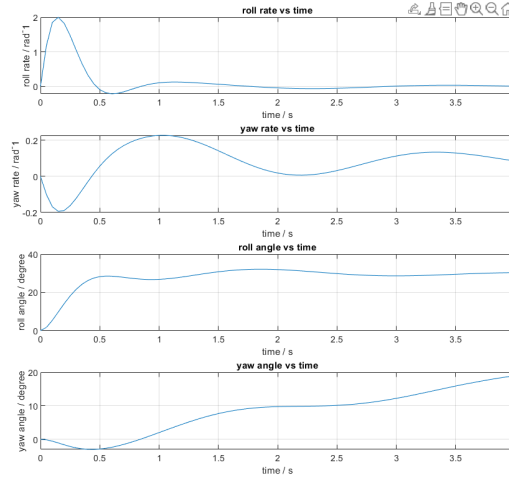


Figure 7: Aircraft Performance with Controller for a 30° Coll by Simulink

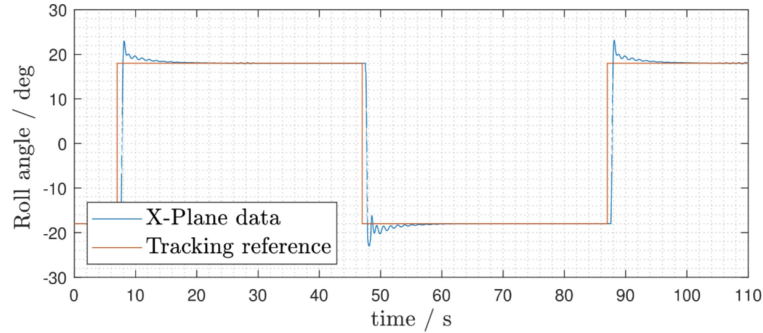


Figure 8: Aircraft Rolling Performance by X-Plane

The yaw controller are purposed to have two part: adverse yaw controller and yaw angle controller. From Figure 6, the yaw rate and yaw angle is positive and have reasonable values for turning when using rolling controller only. As a result, there is no concerns about the adverse yaw, so that the adverse yaw controller is not necessary. Although there is no need for yawing control, a yawing control is still created for potential yawing requirement, which is also successful in Simulink and X-Plane.

4.3 Longitudinal Model

The longitudinal model used is shown in the figure below:

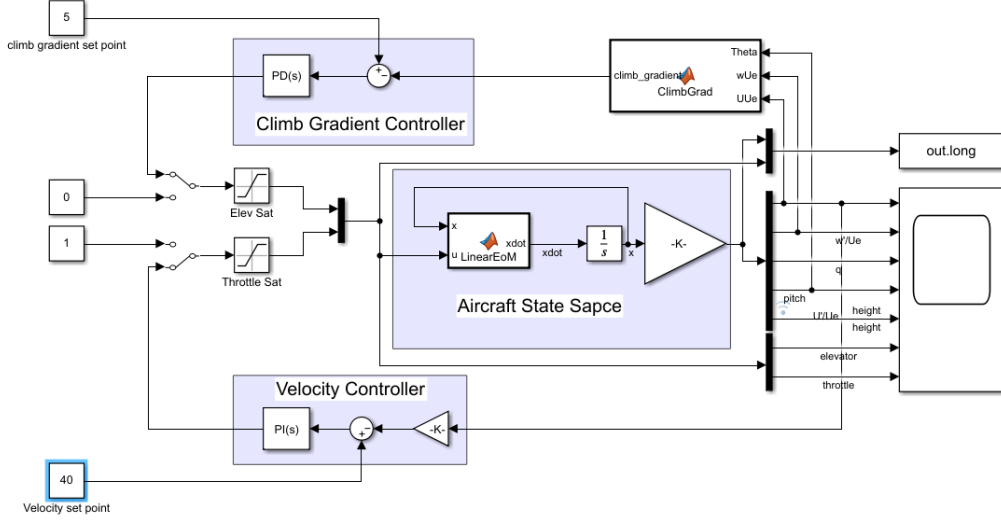


Figure 9: Longitudinal Model

Where the state space formula produces states x , which are u and w velocity, pitch rate, pitch angle and aircraft height. The input u is elevator deflection and throttle in percentage of their maximum

In the altitude control, an climb gradient controller is designed. Through calculating difference between set point and actual climb gradient, the correspond elevator deflection can be conducted trough controller. the climb gradient was calculated through the equation below:

$$\gamma' = \Theta' - \tan^{-1} \left(\frac{W'}{U_e + U'} \right) \quad (30)$$

As for the velocity control, the PID controller produce throttle comment from difference in velocity.

4.4 Longitudinal Control performance

Overall, the longitudinal controller is much harder to tune comparing with lateral ones. And the elevator control derivatives is very large, which suggested that there is over design of the horizon tailplane. An I term is decided to avoid to use, with the same reason from rolling controller. After tuning, the performance of gradient climb is shown below:

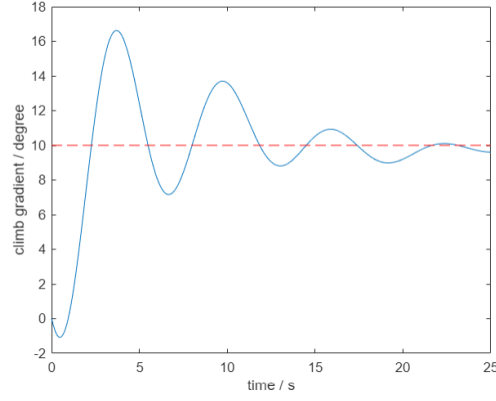


Figure 10: Performance of Climb Gradient at 10°

It can be found that the climb gradient control performance is very oscillatory with long settling time even with tuned controller, which suggested the elevator is over designed, or the control derivatives was wrong. By referring to performance from flight simulation in Figure 11 below, the problem still exists.

As a result, a further investigation in the climb gradient controller will be carried out if time allows, including recalculating and sizing the elevators and their control derivatives, a pitch controller can also be added to reduce oscillation.

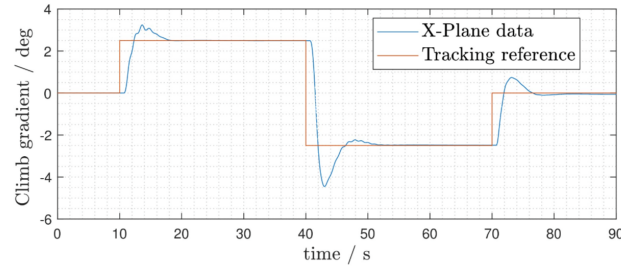


Figure 11: performance of Climb gradient with X-Plane

Lastly, the performance of velocity controller is shown:

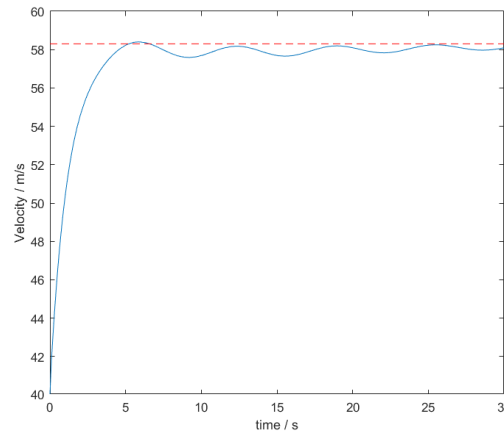


Figure 12: Performance of Velocity Controller from 40m/s to 58.3m/s

Where an PID controller was used. However, there till have a small steady state error and oscillatory behavior.

4.5 Final Control Gains

Controller	K_p	K_i	K_d
Roll	-0.02	-	-0.003
Yaw	0.05	0.001	0.05
Climb Gradient	-0.0025	-	0.005
Velocity	0.3	0.02	0.3

Table 3: Control Derivatives Values

5 Conclusions and Futher Work Recommendations

To conclude, in the aileron design, the it is designed to extend from from 40.91% of span to 63.6%, with chord ratio 0.3 and max deflection $\pm 15^\circ$. Through estimation, it can do a 30° roll within 0.95 second, while the span of aileron is constrained by the stall position and pylon. Further work could be done in finding difference in deflection to eliminate adverse yaw effect under different situation eve though it is small.

As for control derivatives calculation, the rudder and aileron ones are calculated from ESDU while elevator and throttle are based on lecture note derivatives. It clearly shows that those from ESDU is more reasonable in terms of both magnitude and control-ability in the model. More research should be done in finding suitable control derivatives especially the ones fit in this design, as the UAV has an U-tail and with small cruise speed.

As for the autopilot design, control gains for roll, yaw, velocity and climb gradient is conducted. And clearly that lateral ones have better performance. As a result, more controller, for example pitch rate controller should be applied and. Despite the controller, the effect of weather condition are also recommend for further research, together with navigation system and target chasing, eg vessels.

References

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Appendix

TABLE A1.1 CONVERSION FACTORS FOR AERODYNAMIC DERIVATIVES

<i>Dimensional form of derivative</i> (1)	<i>Divisor for forming aero-normalised derivative</i> (2)	<i>Aero-normalised derivative</i> (1)/(2)	<i>Factor for obtaining dynamic-normalised concise derivative</i> (3)	<i>Dynamic-normalised concise derivative</i> (3) × (1)/(2)
$\begin{matrix} \dot{L}_p & \dot{M}_q & \dot{N}_p \\ \dot{L}_r & & \dot{N}_r \end{matrix}$	$\frac{1}{2}\rho_e V_e S l_0^2$	$\begin{matrix} L_p & M_q & N_p \\ L_r & & N_r \end{matrix}$	$\begin{matrix} -1/i_x & & -1/i_z \\ & -1/i_y & \\ -1/i_x & & -1/i_z \end{matrix}$	$\begin{matrix} l_p & m_q & n_p \\ l_r & & n_r \end{matrix}$
$\begin{matrix} \dot{L}_v & \dot{M}_u & \dot{N}_v \\ & \dot{M}_w & \end{matrix}$	$\frac{1}{2}\rho_e V_e S l_0$	$\begin{matrix} L_v & M_u & N_v \\ & M_w & \end{matrix}$	$\begin{matrix} & -\mu/i_y & -\mu/i_z \\ -\mu/i_x & & \\ & -\mu/i_y & \end{matrix}$	$\begin{matrix} l_v & m_u & n_v \\ & m_w & \end{matrix}$
$\dot{M}_{\dot{w}}$	$\frac{1}{2}\rho_e S l_0^2$	$M_{\dot{w}}$	$-1/i_y$	$m_{\dot{w}}$
$\begin{matrix} \dot{L}_\zeta & \dot{M}_\eta & \dot{N}_\zeta \\ \dot{L}_\xi & & \dot{N}_\xi \end{matrix}$	$\frac{1}{2}\rho_e V_e^2 S l_0$	$\begin{matrix} L_\zeta & M_\eta & N_\zeta \\ L_\xi & & N_\xi \end{matrix}$	$\begin{matrix} -\mu/i_x & & -\mu/i_z \\ & -\mu/i_y & \\ -\mu/i_x & & -\mu/i_z \end{matrix}$	$\begin{matrix} l_\zeta & m_\eta & n_\zeta \\ l_\xi & & n_\xi \end{matrix}$
$\begin{matrix} \dot{X}_q & \dot{Y}_p & \dot{Z}_q \\ & \dot{Y}_r & \end{matrix}$	$\frac{1}{2}\rho_e V_e S l_0$	$\begin{matrix} X_q & Y_p & Z_q \\ & Y_r & \end{matrix}$	$\begin{matrix} & -1/\mu & \\ -1/\mu & & -1/\mu \\ & -1/\mu & \end{matrix}$	$\begin{matrix} x_q & y_p & z_q \\ & y_r & \end{matrix}$
$\begin{matrix} \dot{X}_u & \dot{Y}_v & \dot{Z}_u \\ \dot{X}_w & & \dot{Z}_w \end{matrix}$	$\frac{1}{2}\rho_e V_e S$	$\begin{matrix} X_u & Y_v & Z_u \\ X_w & & Z_w \end{matrix}$	$\begin{matrix} -1 & & -1 \\ & -1 & \\ -1 & & -1 \end{matrix}$	$\begin{matrix} x_u & y_v & z_u \\ x_w & & z_w \end{matrix}$
$\dot{X}_{\dot{w}} \quad \dot{Z}_{\dot{w}}$	$\frac{1}{2}\rho_e S l_0$	$X_{\dot{w}} \quad Z_{\dot{w}}$	$-1/\mu \quad -1/\mu$	$x_{\dot{w}} \quad z_{\dot{w}}$
$\begin{matrix} \dot{X}_\eta & \dot{Z}_\eta \\ & \dot{Y}_\zeta \\ & \dot{Y}_\xi \end{matrix}$	$\frac{1}{2}\rho_e V_e^2 S$	$\begin{matrix} X_\eta & Z_\eta \\ & Y_\zeta \\ & Y_\xi \end{matrix}$	$\begin{matrix} -1 & & -1 \\ & -1 & \\ & -1 & \end{matrix}$	$\begin{matrix} x_\eta & z_\eta \\ & y_\zeta \\ & y_\xi \end{matrix}$
$\mu = m_e / \frac{1}{2}\rho_e S l_0$ $i_x, i_y, i_z = I_x / m_e l_0^2, I_y / m_e l_0^2, I_z / m_e l_0^2$				

Figure A : Conventional Factors for Aerodynamics Derivatives